WORKED SOLUTION

STEP II 2005, Question 2

Pure • Euler's Totient Function

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q2

The key idea

Everything follows from writing N in terms of its prime factorisation. Each factor $(1-\frac{1}{p}) = \frac{p-1}{p}$ has a p in the denominator, and N certainly contains at least one p to cancel it. Once $\mathrm{f}(N)$ is written as a product over prime powers, you can see exactly when $\mathrm{f}(m)\mathrm{f}(n)$ and $\mathrm{f}(mn)$ agree: it depends on whether m and n share a prime factor.

Solution

Part (i)

(a) The primes dividing $12 = 2^2 \cdot 3$ are 2 and 3, so

$$\mathrm{f}(12) = 12\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4.$$

The primes dividing $180 = 2^2 \cdot 3^2 \cdot 5$ are 2, 3 and 5, so

$$\mathrm{f}(180) = 180 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 48.$$

(b) For $N = 1$ there are no prime factors, the product is empty, and $\mathrm{f}(1) = 1$. For $N \geq 2$ write

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k},$$

where the p_i are the distinct primes dividing N and every $a_i \geq 1$. Then

$$\begin{aligned}\mathrm{f}(N) &= p_1^{a_1} \cdots p_k^{a_k} \cdot \frac{p_1-1}{p_1} \cdots \frac{p_k-1}{p_k} \\ &= p_1^{a_1-1}(p_1-1)\, p_2^{a_2-1}(p_2-1) \cdots p_k^{a_k-1}(p_k-1).\end{aligned}$$

Since each $a_i \geq 1$, every exponent $a_i - 1$ is a non-negative integer, so each factor $p_i^{a_i-1}(p_i-1)$ is a positive integer. Hence $\mathrm{f}(N)$ is an integer for all N. We will use this

formula,

$$\mathrm{f}(N) = \prod_{i=1}^{k} p_i^{a_i - 1}(p_i - 1), \qquad (*)$$

for the rest of the question.

Part (ii)

(a) This is false. Take $m = n = 2$. Then $\mathrm{f}(2) = 2 \cdot \frac{1}{2} = 1$, so $\mathrm{f}(2)\mathrm{f}(2) = 1$, but $\mathrm{f}(4) = 4 \cdot \frac{1}{2} = 2$.

The problem is that the factor $(1 - \frac{1}{2})$ appears twice on the left but only once on the right, because 2 is counted only once as a prime factor of 4.

(b) This is true. If p and q are distinct primes, then $\mathrm{f}(p) = p(1 - \frac{1}{p}) = p - 1$, and similarly $\mathrm{f}(q) = q - 1$. The primes dividing pq are exactly p and q, and they are different, so

$$\mathrm{f}(pq) = pq\left(1 - \frac{1}{p}\right)\left(1 - \frac{1}{q}\right) = (p-1)(q-1) = \mathrm{f}(p)\mathrm{f}(q).$$

(c) The claim is that $\mathrm{f}(p)\mathrm{f}(q) = \mathrm{f}(pq)$ can hold *only* when p and q are distinct primes; that is, the equation forces p and q to be distinct primes. This is false. Take $p = 3$ and $q = 4$, neither of which needs to be prime for the equation to make sense. Then

$$\mathrm{f}(3)\mathrm{f}(4) = 2 \cdot 2 = 4 = \mathrm{f}(12),$$

by part (i)(a), but 4 is not prime. (An even simpler example is $p = 1$, since $\mathrm{f}(1)\mathrm{f}(q) = \mathrm{f}(q)$ for every q.)

? **Ask yourself** What really makes the equation hold, if not primality?

What matters is that p and q share no prime factor. If $\gcd(m, n) = 1$, then the prime powers in m and in n are different, so the factorisation of mn is just the two factorisations side by side, and $(*)$ gives $\mathrm{f}(mn) = \mathrm{f}(m)\mathrm{f}(n)$. If some prime r divides both, then the factor $(1 - \frac{1}{r})$ appears twice in $\mathrm{f}(m)\mathrm{f}(n)$ but once in $\mathrm{f}(mn)$, so $\mathrm{f}(m)\mathrm{f}(n) < \mathrm{f}(mn)$. So the equation holds exactly when $\gcd(m, n) = 1$, and 3, 4 is just one coprime pair that is not a pair of primes.

If instead you read p and q as standing for primes throughout part (ii), then (c) only asks whether $p = q$ can occur, and it cannot: $\mathrm{f}(p)^2 = (p-1)^2$, while $\mathrm{f}(p^2) = p(p-1)$, and these are never equal because $p - 1 \neq p$. Under that reading (c) is true. The general reading, with the counterexample above, is the more natural one.

Part (iii)

By $(*)$, a prime power has

$$\mathrm{f}(p^m) = p^{m-1}(p-1).$$

Factorising the target gives $146410 = 10 \times 14641 = 2 \cdot 5 \cdot 11^4$, because $14641 = 121^2 = 11^4$. This already has the form $p^{m-1}(p-1)$ with $p = 11$: $11^4 \cdot 10 = 11^4(11-1)$, so $m - 1 = 4$. Hence

$$p = 11, \quad m = 5: \qquad \mathrm{f}(11^5) = 11^4 \cdot 10 = 146410.$$

This is in fact the only answer. If $m \geq 2$, then p divides 146410, so $p \in \{2, 5, 11\}$. Taking $p = 2$ would give $2^{m-1} = 146410$, impossible because 146410 has odd prime factors. Taking $p = 5$ would give $4 \cdot 5^{m-1} = 146410$, impossible because 146410 is not divisible by 4. That leaves $p = 11$, which gives $m = 5$. If $m = 1$ we would need $p = 146411$ to be prime, but $146411 = 41 \times 3571$.

Checks

- The function is Euler's totient: $\mathrm{f}(N)$ counts the integers in $\{1, \ldots, N\}$ coprime to N. For example, $1, 5, 7, 11$ are the four such numbers for $N = 12$. A computer confirmed that the product formula agrees with this count, and is an integer, for every $N < 2000$.
- The worked example in the question gives $\mathrm{f}(80) = 80 \cdot \frac{1}{2} \cdot \frac{4}{5} = 32 = 2^3(2-1) \cdot 5^0(5-1)$, which matches $(*)$.
- The computer tested 20 000 random pairs $m, n \leq 400$ and found $\mathrm{f}(m)\mathrm{f}(n) = \mathrm{f}(mn)$ exactly when $\gcd(m, n) = 1$. A search over all primes $p \leq 146411$ found $(p, m) = (11, 5)$ as the only solution to (iii).

Common slips

- Including a prime more than once in the product: for $180 = 2^2 \cdot 3^2 \cdot 5$ there are three factors, not five.
- Treating "only if" as "if". Part (b) is the "if" direction and part (c) is the converse, which is a different statement.
- Missing that $14641 = 11^4$ in (iii). Spotting $11^2 = 121$ and $121^2 = 14641$, or recognising the row $1, 4, 6, 4, 1$ of Pascal's triangle, finds it quickly.

Beyond STEP

This function is Euler's totient function φ, and the facts in part (ii) say that it is *multiplicative*: $\varphi(mn) = \varphi(m)\varphi(n)$ whenever $\gcd(m, n) = 1$, though not in general. It appears in Euler's theorem, $a^{\varphi(n)} \equiv 1 \pmod{n}$ for a coprime to n, which generalises Fermat's little theorem and underpins RSA encryption. In group theory, $\varphi(n)$ is the order of the group of units modulo n. Davenport's *The Higher Arithmetic* develops all of this beautifully.

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